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Dec. 28, 2021 (KR) 10-2021-0190221

- (51) **Int. Cl.**
H10K 59/88 (2023.01)
G06F 3/041 (2006.01)

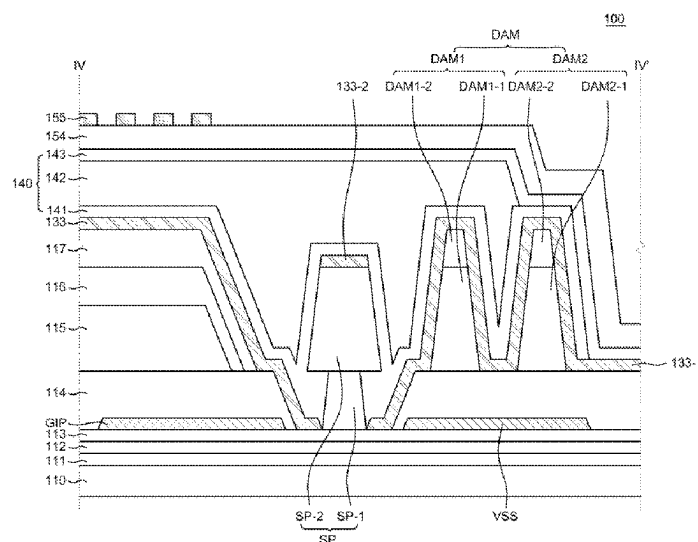
(Continued)

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CPC *H10K 59/88* (2023.02); *G06F 3/0412*
(2013.01); *G06F 3/04164* (2019.05);
(Continued)

- (58) **Field of Classification Search**
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H10K 59/88; H10K 59/131; H10K
50/844

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FIG. 1

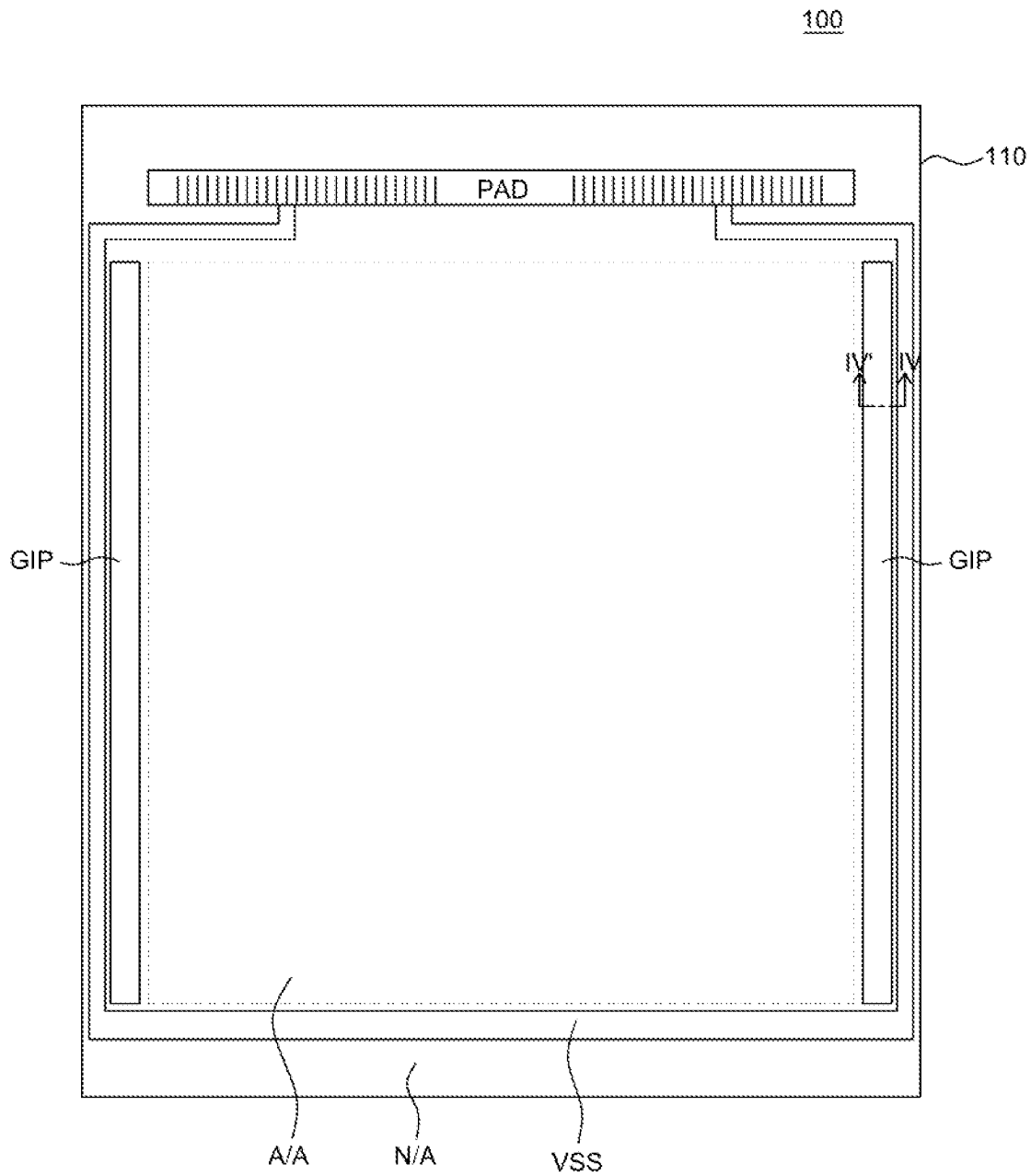


FIG. 2

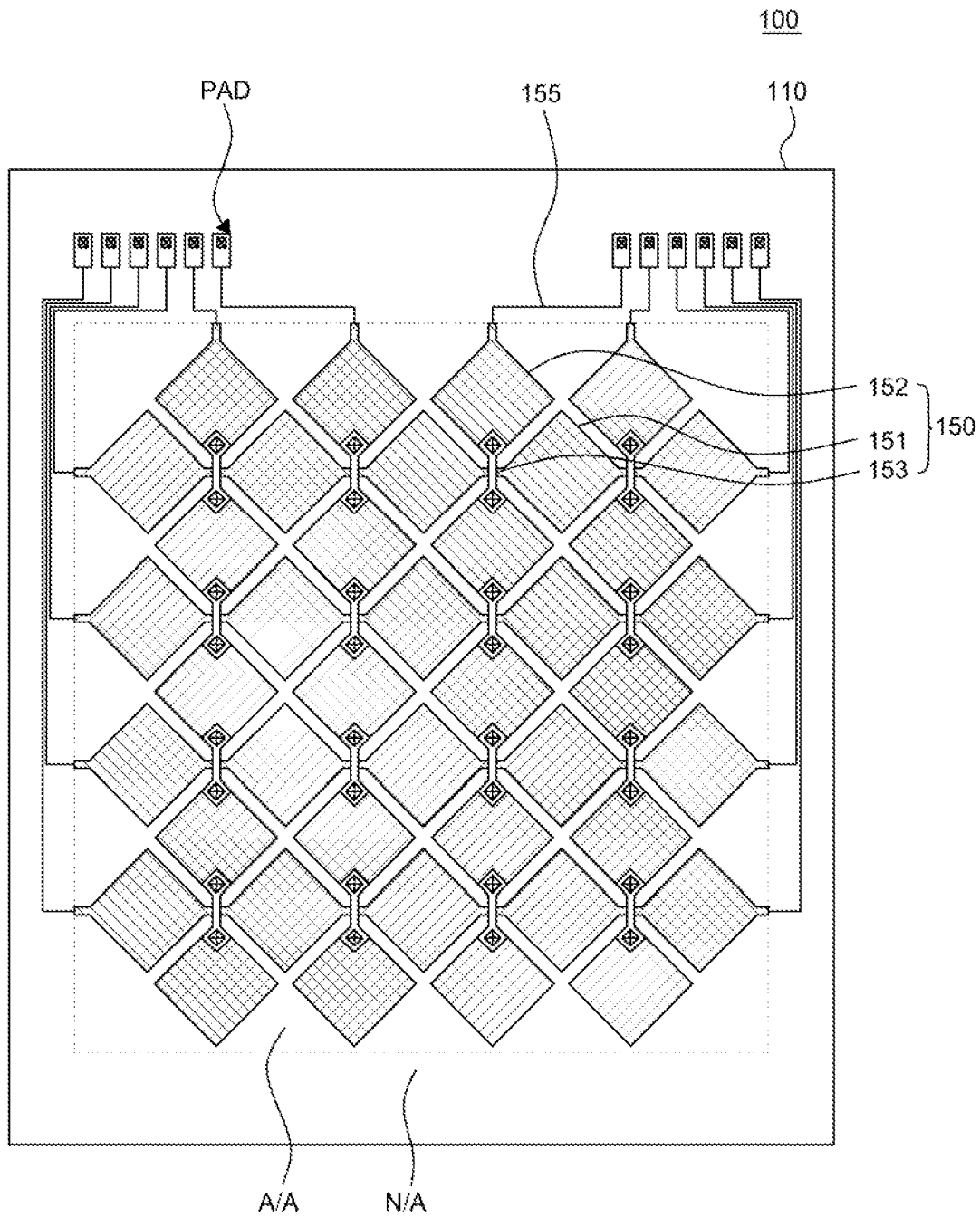


FIG. 3

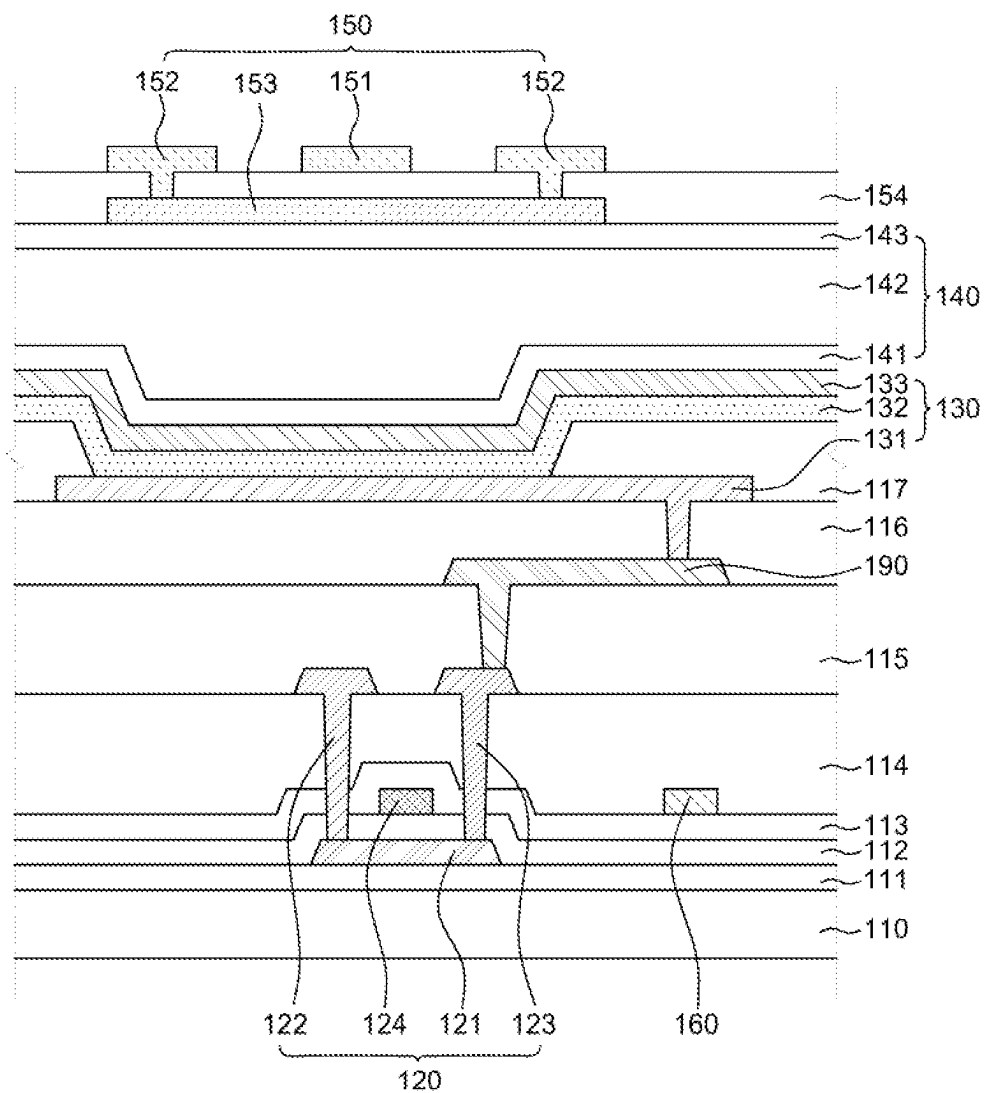


FIG. 4

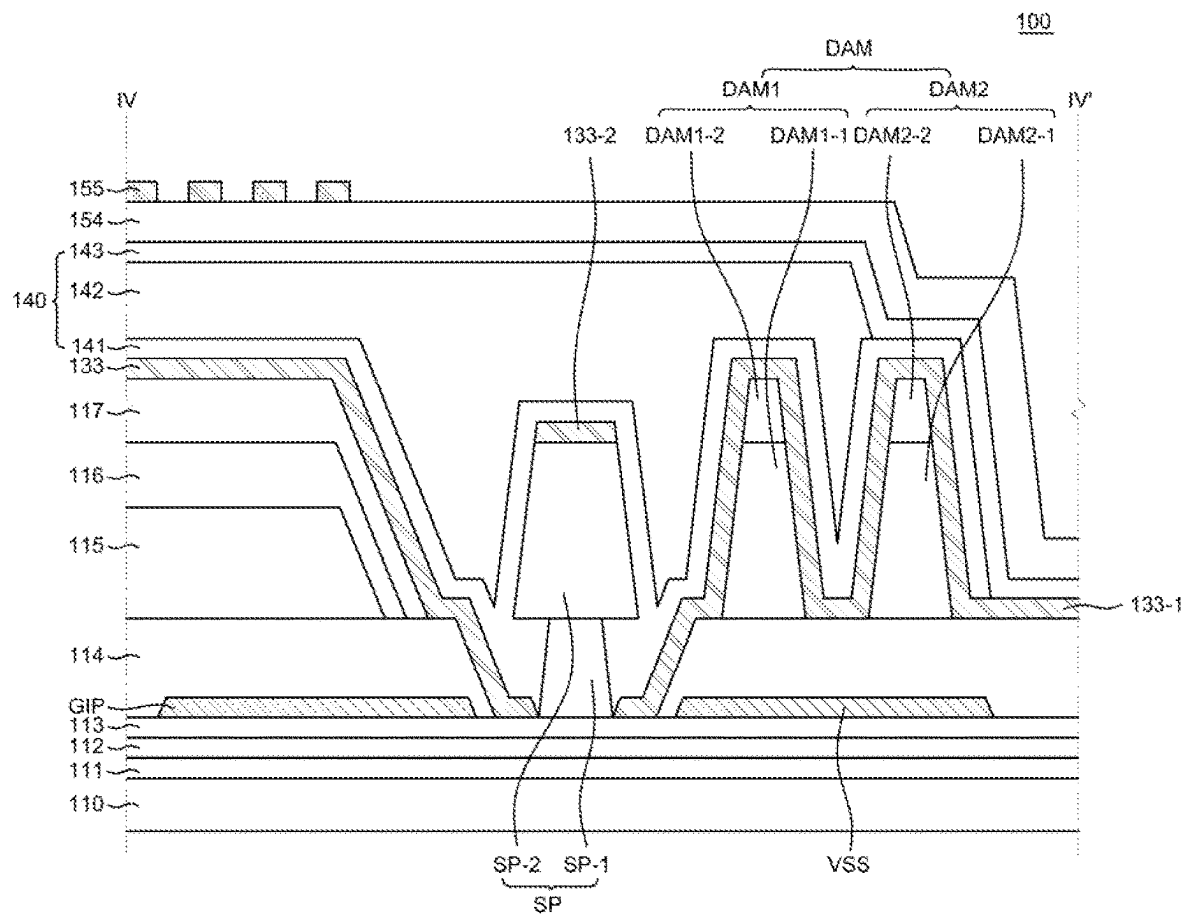
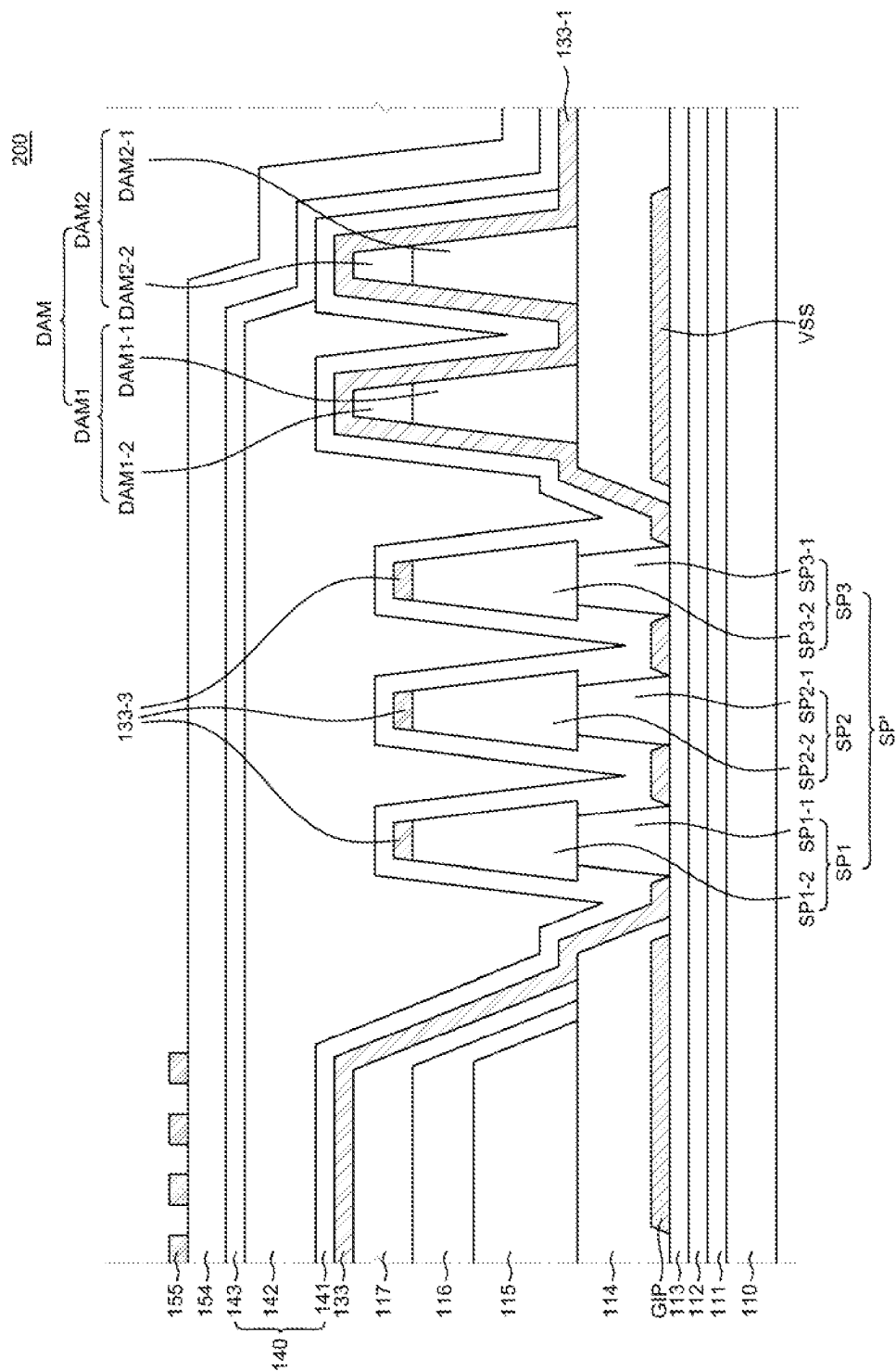


FIG. 5



1

TOUCH DISPLAY DEVICE FOR BLOCKING THE PENETRATION OF MOISTURE INTO THE INTERIOR WHILE EXTENDING THE CATHODE ELECTRODE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/942,200, filed on Sep. 12, 2022, which claims the priority of Korean Patent Application No. 10-2021-0190221 filed on Dec. 28, 2021, which are hereby incorporated by reference in their entirety.

BACKGROUND

Field of the Disclosure

The present disclosure relates to a display device, and more particularly, to a display device capable of improving reliability.

Description of the Background

As the information society has been developed, the demand for a display device that displays an image has increased, and various types of display devices such as a liquid crystal display device and an organic light emitting display device are used.

These display devices provide a function of recognizing a user's touch on a display panel and performing input processing based on the recognized touch to provide more various functions to the user.

The display device capable of recognizing the touch includes a plurality of touch electrodes disposed or embedded in the display panel, and may detect the presence or absence of a user's touch on the display panel, touch coordinates, and the like by driving the touch electrodes.

SUMMARY

Accordingly, the present disclosure is to provide a display device capable of reducing the noise of a touch sensing signal due to signal interference between a gate driving circuit and a touch routing line.

The present disclosure is also to provide a display device capable of blocking the penetration of moisture into the display device while extending a cathode electrode.

Further, the present disclosure is to provide a display device capable of suppressing touch performance from being deteriorated to enable accurate touch sensing.

The present disclosure is not limited to the above-mentioned and other features, which are not mentioned above, may be clearly understood by those skilled in the art from the following descriptions.

According to an aspect of the present disclosure, a display device includes a substrate including a display area and a non-display area surrounding the display area, a gate driving circuit disposed on the substrate and in the non-display area, a light emitting element disposed in the display area on the substrate and including a cathode electrode, a dam disposed on the substrate and in the non-display area, the dam further outside the substrate than the gate driving circuit, a first dummy conductive layer disposed to cover the dam and made of the same material as the cathode electrode, and a spacer disposed between the gate driving circuit and the dam

2

on the substrate, in which the cathode electrode extends from the display area to the non-display area to overlap with the gate driving circuit.

According to another aspect of the present disclosure, a display device includes a substrate including a display area and a non-display area surrounding the display area, a gate driving circuit disposed on the substrate and in the non-display area, a light emitting element disposed in the display area on the substrate and including a cathode electrode, a dam disposed on the substrate and in the non-display area, the dam further outside the substrate than the gate driving circuit, a first dummy conductive layer disposed to cover the dam and made of the same material as the cathode electrode, a plurality of spacers disposed between the gate driving circuit and the dam on the substrate, and a second dummy conductive layer disposed on the plurality of spacers and made of the same material as the cathode electrode, in which the cathode electrode extends from the display area to the non-display area to overlap with the gate driving circuit.

Other detailed matters of the exemplary aspects are included in the detailed description and the drawings.

According to the present disclosure, moisture from penetrating into the display device may be suppressed and reliability by disposing the spacer between the dam and the display area to disconnect the extended cathode electrode may be improved.

According to the present disclosure, the occurrence of interference between the driving signal and the touch signal by extending the cathode electrode may be minimized.

According to the present disclosure, touch precision by reducing noise capable of reaching at touch sensing unit may be improved.

The effects according to the present disclosure are not limited to the contents exemplified above, and more various effects are included in the present specification.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a schematic plan view of a display device according to an exemplary aspect of the present disclosure;

FIG. 2 is a schematic plan view of a display device according to an exemplary aspect of the present disclosure;

FIG. 3 is a cross-sectional view of a display device according to an exemplary aspect of the present disclosure;

FIG. 4 is a cross-sectional view taken along line IV-IV' of FIG. 1; and

FIG. 5 is a cross-sectional view of a display device according to another exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

Advantages and characteristics of the present disclosure and a method of achieving the advantages and characteristics will be clear by referring to exemplary aspects described below in detail together with the accompanying drawings. However, the present disclosure is not limited to the exemplary aspects disclosed herein but will be implemented in various forms. The exemplary aspects are provided by way of example only so that those skilled in the art may fully understand the disclosures of the present disclosure and the

scope of the present disclosure. Therefore, the present disclosure will be defined only by the scope of the appended claims.

The shapes, sizes, ratios, angles, numbers, and the like illustrated in the accompanying drawings for describing the exemplary aspects of the present disclosure are merely examples, and the present disclosure is not limited thereto. Like reference numerals generally denote like elements throughout the specification. Further, in the following description of the present disclosure, a detailed explanation of known related technologies may be omitted to avoid unnecessarily obscuring the subject matter of the present disclosure. The terms such as “including,” “having,” and “consist of” used herein are generally intended to allow other components to be added unless the terms are used with the term “only”. Any references to singular may include plural unless expressly stated otherwise.

Components are interpreted to include an ordinary error range even if not expressly stated.

When the position relation between two parts is described using the terms such as “on”, “above”, “below”, and “next”, one or more parts may be positioned between the two parts unless the terms are used with the term “immediately” or “directly”.

When an element or layer is disposed “on” another element or layer, another layer or another element may be interposed directly on the other element or therebetween.

Although the terms “first”, “second”, and the like are used for describing various components, these components are not confined by these terms. These terms are merely used for distinguishing one component from the other components. Therefore, a first component to be mentioned below may be a second component in a technical concept of the present disclosure.

Like reference numerals generally denote like elements throughout the specification.

A size and a thickness of each component illustrated in the drawing are illustrated for convenience of description, and the present disclosure is not limited to the size and the thickness of the component illustrated.

The features of various aspects of the present disclosure may be partially or entirely adhered to or combined with each other and may be interlocked and operated in technically various ways, and the aspects may be carried out independently of or in association with each other.

Hereinafter, a display device according to exemplary aspects of the present disclosure will be described in detail with reference to accompanying drawings.

FIG. 1 is a schematic plan view of a display device according to an exemplary aspect of the present disclosure. In FIG. 1, for convenience of description, among various components of a display device **100**, a substrate **110**, a pad PAD, a gate driving circuit GIP, and a low potential voltage line VSS are just illustrated.

Referring to FIG. 1, the display device **100** includes at least one display area A/A, and an array of pixels is formed in the display area A/A. A non-display area N/A may be disposed to surround the display area A/A. That is, the non-display area N/A may be adjacent to one or more side surfaces of the display area A/A. In FIG. 1, the non-display area N/A surrounds a quadrangular display area A/A. However, the shape of the display area A/A and the shape/disposition of the non-display area N/A adjacent to the display area A/A are not limited to the example illustrated in FIG. 1.

Each pixel in the display area A/A may be associated with a pixel circuit. The pixel circuit may include one or more

switching transistors and one or more driving transistors on a backplane. Each pixel circuit may be electrically connected to a gate line and a data line to communicate with one or more driving circuits such as a gate driving circuit GIP and a data driving circuit located in the non-display area N/A. In addition, each pixel circuit may be connected to power supply lines VDD, VSS, VREF, etc. located in the non-display area N/A to receive a voltage required for driving. The driving circuit may be implemented as a thin film transistor (TFT) in the non-display area N/A, as illustrated in FIG. 1. Such a driving circuit may be referred to as a gate-in-panel (GIP). In addition, some components, such as a data driver IC, are mounted on a separate printed circuit board and may be coupled to a connection interface (PAD, bump, pin, etc.) disposed in the non-display area N/A using circuit films such as a flexible printed circuit board (FPCB), a chip-on-film (COF), and a tape carrier-package (TCP). The gate driving circuit GIP may be disposed on at least one side of the left and right sides of the display area A/A.

A power supply unit of the display device **100** outputs a high potential voltage, a low potential voltage, and the like. The high potential voltage, the low potential voltage, and the like output from the power supply unit are supplied to the display device **100**. The high potential voltage is supplied to the display panel through a high potential voltage line, and the low potential voltage is supplied to the display device **100** through a low potential voltage line VSS. The voltage output from the power supply unit is also used in the data driving circuit or the gate driving circuit GIP. In particular, the low potential voltage line VSS is a line that may be electrically connected to a second electrode of a light emitting element to be described below, and may be disposed to surround the gate driving circuit GIP and the display area A/A.

In addition, the non-display area N/A may include a pad unit, and the pad unit may include a plurality of pads PAD. Specifically, the plurality of pads PAD may include a plurality of common power supply pads, a plurality of data input pads, a plurality of power supply pads, a plurality of control signal input pads, a plurality of touch driving pads, and the like.

The display device **100** may further include various additional elements for generating various signals or driving pixels in the display area A/A. The additional elements for driving the pixels may be an inverter circuit, a multiplexer, an electrostatic discharge circuit, and the like. The display device **100** may also include additional elements associated with functions other than driving pixels. For example, the display device **100** may include additional elements that provide a touch sensing function, a user authentication function (e.g., fingerprint recognition), a multi-level pressure sensing function, a tactile feedback function, and the like. The above-mentioned additional elements may be located in the non-display area N/A and/or in an external circuit connected to the connection interface.

FIG. 2 is a schematic plan view of the display device according to the exemplary aspect of the present disclosure. In FIG. 2, for convenience of description, among various components of the display device **100**, the substrate **110**, the touch sensing unit **150**, the touch routing line **155**, and the plurality of pads PAD are just illustrated.

Recently, to reduce the thickness of the display device and improve visibility, a touch panel-integrated display device in which a touch electrode and the like are disposed directly on an encapsulation unit of the display device has been developed. The display device **100** according to the exemplary aspect of the present disclosure is a touch panel-integrated

display device, and the touch sensing unit **150** may be formed on the encapsulation unit of the display device **100**.

Referring to FIG. 2, the display device **100** includes the touch sensing unit **150**, the plurality of touch routing lines **155**, and the plurality of touch pads PAD. Here, the touch sensing unit **150** includes a plurality of first touch electrodes **151**, a plurality of second touch electrodes **152**, and a connection electrode **153**.

The plurality of first touch electrodes **151** may be touch driving electrodes, and the plurality of second touch electrodes **152** may be touch sensing electrodes. The plurality of first touch electrodes **151** is connected to each other in a row direction to form a plurality of electrode rows, and the plurality of second touch electrodes **152** is connected to each other in a longitudinal direction by the connection electrode **153** to form a plurality of electrode columns.

The first touch electrodes **151** and the second touch electrodes **152** may be disposed on the same layer. However, in a region where the first touch electrode **151** and the second touch electrode **152** intersect, the second touch electrodes **152** are disposed separately, and the second touch electrodes **152** separated from each other may be connected to each other by the connection electrode **153**.

In this case, the first touch electrode **151**, the second touch electrode **152**, and the connection electrode **153** are disposed in an area corresponding to the display area A/A of the display device **100**.

The external shapes of the first touch electrode **151** and the second touch electrode **152** may correspond to specific shapes. For example, as illustrated in FIG. 2, the external shapes of the first touch electrode **151** and the second touch electrode **152** may have mesh patterns including a plurality of rhombus shapes. The first touch electrode **151** and the second touch electrode **152** may be made of metal including at least one of titanium (Ti), aluminum (Al), molybdenum (Mo), molytitanium (MoTi), copper (Cu), and tantalum (Ta) and may be made of a transparent conductive material such as indium tin oxide (ITO) or indium zinc oxide (IZO), but is not limited thereto. Light emitted from the display device **100** may pass through the first touch electrode **151** and the second touch electrode **152** made of the transparent conductive material to be emitted to the outside. However, the present disclosure is not limited thereto, and the light emitted from the display device **100** may also be emitted to the outside through a plurality of openings included in the first touch electrode **151** and the second touch electrode **152**.

The non-display area N/A is an area surrounding the display area A/A, and the plurality of touch routing line **155** and the plurality of touch pads PAD are disposed.

Each of the plurality of touch routing lines **155** electrically connects each of the plurality of touch electrodes **151** and **152** disposed in the display area A/A to the touch pad of the non-display area N/A. For example, a touch driving signal may be applied to the first touch electrode **151** through the touch routing line **155** connected to the first touch electrode **151**, and a touch sensing signal may be transmitted to the second touch electrode **152** through the touch routing line **155** connected to the second touch electrode **152**.

These touch routing lines **155** may be made of a low-resistance metal material, and may also be made of a transparent conductive material such as ITO or IZO, but are not limited thereto. For example, when the plurality of touch routing lines **155** is made of the low-resistance metal material, resistance may be lowered, thereby reducing an RC delay.

One end of the plurality of touch pads PAD is connected to the touch routing line **155**, and the other end thereof is electrically connected to an external circuit such as a touch driving unit to receive a touch signal from the external circuit or to transmit a touch sensing signal to the external circuit.

In this case, the plurality of touch routing lines **155** and the plurality of touch pads PAD are disposed in an area corresponding to the non-display area N/A of the display device **100**.

Hereinafter, a cross-sectional structure of the display area A/A of the display device **100** will be described in more detail with reference to FIG. 3 together.

FIG. 3 is a cross-sectional view of the display device according to the exemplary aspect of the present disclosure.

Referring to FIG. 3, the display device **100** according to the exemplary aspect of the present disclosure may include a substrate **110**, a buffer layer **111**, a first thin film transistor **120**, a gate insulating layer **112**, a first interlayer insulating layer **113**, a conductive layer **160**, a second interlayer insulating layer **114**, a first planarization layer **115**, a connection electrode **190**, a second planarization layer **116**, a bank **117**, a light emitting element **130**, an encapsulation unit **140**, a touch insulating layer **154**, and a touch sensing unit **150**.

The substrate **110** may support various components of the display device **100**. The substrate **110** may be made of glass or a plastic material having flexibility. When the substrate **110** is made of the plastic material, the substrate **110** may be made of, for example, polyimide (PI).

The buffer layer **111** may be disposed on the substrate **110**. The buffer layer **111** may be formed of a single layer of silicon nitride (SiNx) or silicon oxide (SiOx) or a multilayer thereof. The buffer layer **111** may serve to enhance adhesion between layers formed on the buffer layer **111** and the substrate **110** and block an alkali component, and the like which leaks from the substrate **110**.

The thin film transistor **120** may be disposed on the buffer layer **111**. The thin film transistor **120** includes an active layer **121**, a gate electrode **124**, a source electrode **122**, and a drain electrode **123**. Here, depending on a design of the pixel circuit, the source electrode **122** may be a drain electrode, and the drain electrode **123** may be a source electrode. The active layer **121** of the thin film transistor **120** may be disposed on the buffer layer **111**.

The active layer **121** may be made of various materials such as polysilicon, amorphous silicon, and oxide semiconductor. The active layer **121** may include a channel region in which a channel is formed when the thin film transistor **120** is driven, and a source region and a drain region on both sides of the channel region. The source region refers to a portion of the active layer **121** connected to the source electrode **122**, and the drain region refers to a portion of the active layer **121** connected to the drain electrode **123**.

The gate insulating layer **112** may be disposed on the active layer **121** of the thin film transistor **120**. The gate insulating layer **112** may be formed of a single layer of silicon nitride (SiNx) or silicon oxide (SiOx) or a multilayer thereof. The gate insulating layer **112** has a contact hole through which the source electrode **122** and the drain electrode **123** of the thin film transistor **120** are connected to the source region and the drain region of the active layer **121** of the thin film transistor **120**, respectively.

The gate electrode **124** of the thin film transistor **120** may be disposed on the gate insulating layer **112**. The gate electrode **124** may be formed of a single layer or a multilayer made of any one of molybdenum (Mo), copper (Cu), tita-

nium (Ti), aluminum (Al), chromium (Cr), gold (Au), nickel (Ni), and neodymium (Nd), or alloys thereof. The gate electrode **124** may be formed on the gate insulating layer **112** to overlap with the channel region of the active layer **121** of the thin film transistor **120**.

The first interlayer insulating layer **113** may be disposed on the gate insulating layer **112** and the gate electrode **124**. The first interlayer insulating layer **113** may be formed of a single layer of silicon nitride (SiNx) or silicon oxide (SiOx) or a multilayer thereof. A contact hole for exposing the source region and the drain region of the active layer **121** of the thin film transistor **120** may be formed in the first interlayer insulating layer **113**.

The conductive layer **160** may be disposed on the first interlayer insulating layer **113**. The conductive layer **160** may be lines or electrodes disposed among the gate electrode **124**, the source electrode **122** and the drain electrode **123**.

The second interlayer insulating layer **114** may be disposed on the first interlayer insulating layer **113** and the conductive layer **160**. The second interlayer insulating layer **114** may be formed of the same material as the first interlayer insulating layer **113**. That is, the second interlayer insulating layer **114** may be formed of a single layer of silicon nitride (SiNx) or silicon oxide (SiOx) or a multilayer thereof. A contact hole for exposing the source region and the drain region of the active layer **121** of the thin film transistor **120** may be formed in the second interlayer insulating layer **114**.

The source electrode **122** and the drain electrode **123** of the thin film transistor **120** may be disposed on the second interlayer insulating layer **114**.

The source electrode **122** and the drain electrode **123** of the thin film transistor **120** may be connected to the active layer **121** of the thin film transistor **120** through contact holes formed in the gate insulating layer **112**, the first interlayer insulating layer **113**, and the second interlayer insulating layer **114**. Accordingly, the source electrode **122** of the thin film transistor **120** may be connected to the source region of the active layer **121** through the contact holes formed in the gate insulating layer **112**, the first interlayer insulating layer **113**, and the second interlayer insulating layer **114**. In addition, the drain electrode **123** of the thin film transistor **120** may be connected to the drain region of the active layer **121** through the contact holes formed in the gate insulating layer **112**, the first interlayer insulating layer **113**, and the second interlayer insulating layer **114**.

The source electrode **122** and the drain electrode **123** of the thin film transistor **120** may be formed by the same process. In addition, the source electrode **122** and the drain electrode **123** of the thin film transistor **120** may be formed of the same material. The source electrode **122** and the drain electrode **123** of the thin film transistor **120** may be formed of a single layer or a multilayer made of any one of molybdenum (Mo), copper (Cu), titanium (Ti), aluminum (Al), chromium (Cr), gold (Au), nickel (Ni), and neodymium (Nd), or alloys thereof.

The first planarization layer **115** may be disposed on the source electrode **122**, the drain electrode **123**, and the second interlayer insulating layer **114**. As illustrated in FIG. 3, a contact hole for exposing the drain electrode **123** may be formed in the first planarization layer **115**. The first planarization layer **115** may be an organic material layer for planarizing an upper portion of the thin film transistor **120**. For example, the first planarization layer **115** may be formed of an organic material, such as an acryl resin, an epoxy resin, a phenolic resin, a polyamide resin, a polyimide resin, etc.

However, the present disclosure is not limited thereto, and the first planarization layer **115** may be an inorganic material layer for protecting the thin film transistor **120**. For example, the first planarization layer **115** may be formed of an inorganic material such as silicon nitride (SiNx) or silicon oxide (SiOx). The first planarization layer **115** may be formed of a single layer of silicon nitride (SiNx) or silicon oxide (SiOx) or a multilayer thereof.

The connection electrode **190** may be disposed on the first planarization layer **115**. In addition, the connection electrode **190** may be connected to the drain electrode **123** of the thin film transistor **120** through the contact hole of the first planarization layer **115**. The connection electrode **190** may serve to electrically connect the thin film transistor **120** and the light emitting element **130**. For example, the connection electrode **190** may serve to electrically connect the drain electrode **123** of the thin film transistor **120** and the first electrode **131** of the light emitting element **130**. The connection electrode **190** may be formed in a single layer or a multilayer made of any one of molybdenum (Mo), copper (Cu), titanium (Ti), aluminum (Al), chromium (Cr), gold (Au), nickel (Ni), and neodymium (Nd), and/or alloys thereof. The connection electrode **190** may be formed of the same material as the source electrode **122** and the drain electrode **123** of the thin film transistor **120**.

The second planarization layer **116** may be disposed on the connection electrode **190** and the first planarization layer **115**. In addition, as illustrated in FIG. 3, a contact hole for exposing the connection electrode **190** may be formed in the second planarization layer **116**. The second planarization layer **116** may be an organic material layer for planarizing an upper portion of the thin film transistor **120**. For example, the second planarization layer **116** may be formed of an organic material such as an acryl resin, an epoxy resin, a phenolic resin, a polyamide resin, a polyimide resin, etc.

The light emitting element **130** may be disposed on the second planarization layer **116**. The light emitting element **130** may include a first electrode **131**, a light emitting structure **132**, and a second electrode **133**. The first electrode **131** of the light emitting element **130** may be disposed on the second planarization layer **116**. The first electrode **131** may be electrically connected with the connection electrode **190** through the contact hole formed in the second planarization layer **116**. Accordingly, the first electrode **131** of the light emitting element **130** is connected with the connection electrode **190** through the contact hole formed in the second planarization layer **116** to be electrically connected with the thin film transistor **120**.

The first electrode **131** may be formed of a multilayer structure including a transparent conductive film and an opaque conductive film with high reflection efficiency. The transparent conductive layer may be made of a material having a relatively large work function value, such as indium-tin-oxide (ITO) or indium-zinc-oxide (IZO). In addition, the opaque conductive film may be formed in a single layer or a multilayer structure including Al, Ag, Cu, Pb, Mo, Ti, or an alloy thereof. For example, the first electrode **131** may be formed in a structure in which a transparent conductive layer, an opaque conductive layer, and a transparent conductive layer are sequentially stacked. However, the present disclosure is not limited thereto, and the first electrode **131** may be formed even in a structure in which the transparent conductive layer and the opaque conductive layer are sequentially stacked.

Since the display device **100** according to the exemplary aspect of the present disclosure is a top emission display device, the first electrode **131** may be an anode electrode.

When the display device **100** is a bottom emission display device, the first electrode **131** disposed on the second planarization layer **116** may be a cathode electrode.

The bank **117** may be disposed on the first electrode **131** and the second planarization layer **116**. An opening for exposing the first electrode **131** may be formed in the bank **117**. The bank **117** may define a light emitting region of the display device **100**, which may also be referred to as a pixel defining layer.

The light emitting structure **132** including a light emitting layer may be disposed on the first electrode **131**.

The light emitting structure **132** of the light emitting element **130** may be formed by stacking a hole layer, a light emitting layer, and an electron layer on the first electrode **131** in the order or in the reverse order. In addition, the light emitting structure **132** may also include first and second light emitting structures facing each other with a charge generation layer interposed therebetween. In this case, the light emitting layer of any one of the first and second light emitting structures generates blue light, and the light emitting layer of the other structure of the first and second light emitting structures generates yellow-green light, so that white light may be generated through the first and second light emitting structures. The white light generated by the light emitting structure **132** may be incident on a color filter located on the upper portion of the light emitting structure **132** to implement a color image. In addition, a color image may be implemented by generating color light corresponding to each subpixel in each light emitting structure **132** without a separate color filter. For example, the light emitting structure **132** of a red (R) subpixel may generate red light, the light emitting structure **132** of a green (G) subpixel may generate green light, and the light emitting structure **132** of a blue (B) subpixel may generate blue light.

The second electrode **133** may be further disposed on the light emitting structure **132**. The second electrode **133** of the light emitting element **130** may be disposed on the light emitting structure **132** to face the first electrode **131** with the light emitting structure **132** interposed therebetween. In the display device **100** according to the exemplary aspect of the present disclosure, the second electrode **133** may be a cathode electrode. The encapsulation unit **140** for suppressing penetration of moisture may be further disposed on the second electrode **133**.

The encapsulation unit **140** may include a first inorganic encapsulation layer **141**, an organic encapsulation layer **142**, and a second inorganic encapsulation layer **143**. The first inorganic encapsulation layer **141** of the encapsulation unit **140** may be disposed on the second electrode **133**. In addition, the organic encapsulation layer **142** may be disposed on the first inorganic encapsulation layer **141**. In addition, the second inorganic encapsulation layer **143** may be disposed on the organic encapsulation layer **142**. The first inorganic encapsulation layer **141** and the second inorganic encapsulation layer **143** of the encapsulation unit **140** may be formed of an inorganic material, such as silicon nitride (SiNx) or silicon oxide (SiOx). The organic encapsulation layer **142** of the encapsulation unit **140** may be formed of an organic material, such as an acryl resin, an epoxy resin, a phenolic resin, a polyamide resin, a polyimide resin, etc.

The second inorganic encapsulation layer **143** may cover an upper surface and a side surface of each of the first inorganic encapsulation layer **141** and the organic encapsulation layer **142**, and the second inorganic encapsulation layer **143** minimizes or blocks external moisture or oxygen from penetrating into the first inorganic encapsulation layer **141** and the organic encapsulation layer **142**. At this time,

the first inorganic encapsulation layer **141** and the second inorganic encapsulation layer **143** serve to block the penetration of moisture or oxygen, and the organic encapsulation layer **142** serves to planarize the upper portion of the first inorganic encapsulation layer **141**. Accordingly, the encapsulation unit **140** may cover the gate driving circuit GIP and the dam in the display area A/A and the non-display area N/A. However, the configuration of the encapsulation layer **140** is not limited thereto.

The touch sensing unit **150** may be disposed on the second inorganic encapsulation layer **143** of the encapsulation unit **140**. The touch sensing unit **150** may include touch electrodes **151** and **152** and a connection electrode **153**.

The connection electrode **153** of the touch sensing unit **150** may be disposed on the second inorganic encapsulation layer **143**. The connection electrode **153** is disposed at a point where the touch electrodes **151** and **152** disposed in different directions intersect to be used to connect the touch electrodes **151** and **152** disposed in any one direction. The connection electrode **153** may be formed of a transparent conductive layer, for example, a transparent conductive oxide such as ITO or IZO.

A touch insulating layer **154** may be disposed on the encapsulation unit **140** and the connection electrode **153**. The touch insulating layer **154** may insulate the connection electrode **153** and the touch electrodes **151** and **152** from each other. In addition, the touch insulating layer **154** may be formed of an inorganic material layer or an organic material layer. When the touch insulating layer **154** is an inorganic material layer, the touch insulating layer **154** may be formed of a single layer of silicon nitride (SiNx) or silicon oxide (SiOx) or a multilayer thereof. In addition, when the touch insulating layer **154** is an organic material layer, the touch insulating layer **154** may be formed of an organic material, such as an acryl resin, an epoxy resin, a phenolic resin, a polyamide resin, a polyimide resin, etc. A contact hole for exposing the connection electrode **153** may be formed in the touch insulating layer **154**.

A plurality of first touch electrodes **151** and a plurality of second touch electrodes **152** of the touch sensing unit **150** may be disposed on the touch insulating layer **154**. The plurality of second touch electrodes **152** may be connected to the connection electrode **153** through a contact hole of the touch insulating layer **154**. The plurality of second touch electrodes **152** may be connected to each other by the connection electrode **153**.

The first touch electrode **151** and the second touch electrode **152** of the touch sensing unit **150** may be formed of a transparent conductive film such as indium-tin-oxide (ITO) or indium-zinc-oxide (IZO). However, the present disclosure is not limited thereto, and the first touch electrode **151** and the second touch electrode **152** may also be formed of an opaque conductive layer having an opening. When the first touch electrode **151** and the second touch electrode **152** are formed of the opaque conductive layer having the opening, the first touch electrode **151** and the second touch electrode **152** may be formed of a single layer or a multilayer made of any one of molybdenum (Mo), copper (Cu), titanium (Ti), aluminum (Al), chromium (Cr), gold (Au), nickel (Ni), and neodymium (Nd), or alloys thereof.

FIG. 4 is a cross-sectional view taken along line IV-IV' of FIG. 1.

The gate driving circuit GIP, the low potential voltage line VSS, a dam DAM, the touch routing line **155**, and a spacer SP may be disposed in the non-display area N/A of the display device **100**.

11

The gate driving circuit GIP receives a gate control signal through gate control lines, generates gate signals according to the gate control signal, and sequentially outputs the signals to the gate lines of the display area A/A. The gate driving circuit GIP may be disposed in the non-display area N/A of the substrate **110** in a gate driver in panel (GIP) manner. The gate driving circuit GIP may be formed on the same layer as the conductive layer **160** of the display area A/A. That is, the gate driving circuit GIP may be disposed between the first interlayer insulating layer **113** and the second interlayer insulating layer **114**. However, the present disclosure is not limited thereto.

The low potential voltage line VSS may supply a common voltage to the second electrode **133** of the light emitting element **130** formed in each pixel of the display area A/A. Accordingly, the low potential voltage line VSS may be disposed to surround the gate driving circuit GIP and the display area A/A. That is, the low potential voltage line VSS may be disposed in an outer region of the substrate **110** further than the gate driving circuit GIP. The low potential voltage line VSS may be formed on the same layer as the gate driving circuit GIP. That is, the low potential voltage line VSS may be formed on the same layer as the conductive layer **160** of the display area A/A, and may be disposed between the first interlayer insulating layer **113** and the second interlayer insulating layer **114**. However, the present disclosure is not limited thereto.

The dam DAM may be disposed on the first interlayer insulating layer **114** in the non-display area N/A. The dam DAM is disposed to control the spread of the organic encapsulation layer forming the encapsulation unit disposed in the display area A/A and the non-display area N/A of the substrate **110**. The dam DAM may include a first dam DAM1 and a second dam DAM2. The first dam DAM1 may surround the display area A/A adjacent to the display area A/A, and the second dam DAM2 may surround the outside of the first dam DAM1. In addition, as illustrated in FIG. 4, the first dam DAM1 and the second dam DAM2 may be disposed on the outside of the substrate **110** more than the gate driving circuit GIP, and may be disposed on the upper portion of the low potential voltage line VSS.

The first dam DAM1 and the second dam DAM2 may be formed in multiple layers. Specifically, the first dam DAM1 may include a first layer DAM1-1 of the first dam DAM1 and a second layer DAM1-2 of the first dam DAM1. In addition, the second dam DAM2 may include a first layer DAM2-1 of the second dam DAM2 and a second layer DAM2-2 of the second dam DAM2. The first layer DAM1-1 of the first dam DAM1 and the first layer DAM2-1 of the second dam DAM2 may be disposed on the second interlayer insulating layer **114**. The first layer DAM1-1 of the first dam DAM1 and the first layer DAM2-1 of the second dam DAM2 may be formed on the same layer as the second planarization layer **116**, and may be formed of the same material. However, the present disclosure is not limited thereto.

The second layer DAM1-2 of the first dam DAM1 may be disposed on the first layer DAM1-1 of the first dam DAM1. In addition, the second layer DAM2-2 of the second dam DAM2 may be disposed on the first layer DAM2-1 of the second dam DAM2. The second layer DAM1-2 of the first dam DAM1 and the second layer DAM2-2 of the second dam DAM2 may be formed on the same layer as the bank **117** and may be made of the same material.

However, in the exemplary aspect of the present disclosure, although the dam DAM is illustrated as having a

12

double layer, the display device **100** is not limited thereto. For example, the dam DAM may be formed of three or more layers.

Next, the second electrode **133**, that is, the cathode electrode may extend from the display area A/A to the non-display area N/A and may be disposed to overlap with the gate driving circuit GIP. The second electrode **133** may extend to a partial region of the non-display area N/A to be electrically connected to the low potential voltage line VSS disposed in the non-display area N/A. The second electrode **133** may extend along an upper surface of the bank **117**, a side surface of the bank **117**, and the second interlayer insulating layer **114** disposed in the non-display area N/A.

A capping layer may be further disposed on the second electrode **133** by protecting the light emitting element **130** to increase emission efficiency. Although not illustrated in the drawings, the capping layer may be disposed between the second electrode **133** and the first inorganic encapsulation layer **141** of the encapsulation unit **140**. When the second electrode **133** is designed to extend from the display area A/A to the non-display area N/A over the first dam DAM1 and the second dam DAM2, the capping layer may be designed to extend together with the second electrode **133**. However, the present disclosure is not limited thereto.

The encapsulation unit **140** for blocking penetration of moisture and oxygen into the light emitting element **130** may be disposed on the second electrode **133**. In addition, the touch insulating layer **154** may be disposed on the encapsulation unit **140** in the non-display area N/A, and a plurality of touch routing lines **155** may be disposed on the touch insulating layer **154**. The plurality of touch routing lines **155** may be disposed to overlap with the second electrode **133** as illustrated in FIG. 4. Each of the plurality of touch routing lines **155** electrically connects each of the plurality of touch electrodes **151** and **152** disposed in the display area AA to the plurality of pads PAD of the non-display area N/A. For example, a touch driving signal may be applied to the first touch electrode **151** through the touch routing line **155** connected to the first touch electrode **151**, and a touch sensing signal may be transmitted to the second touch electrode **152** through the touch routing line **155** connected to the second touch electrode **152**.

Next, the spacer SP may be disposed between the gate driving circuit GIP and the dam DAM on the substrate **110**. That is, the spacer SP may be disposed between the gate driving circuit GIP and the low potential voltage line VSS. Since the gate driving circuit GIP is disposed on at least one side of a left side and a right side of the display area A/A, the spacer SP may also be disposed on at least one side of the left and right sides of the display area A/A.

The spacer SP may be formed of a multilayer. Specifically, the spacer SP may include a first spacer layer SP-1 and a second spacer layer SP-2. The first spacer layer SP-1 may be made of the same material as the first insulating layer disposed in the display area A/A and the non-display area N/A, and the second spacer layer SP-2 may be formed of the same material as the second insulating layer disposed in the display area A/A and the non-display area N/A. In this case, the first insulating layer disposed in the display area A/A and the non-display area N/A may be an inorganic insulating layer, and the second insulating layer may be an organic insulating layer. Specifically, the first insulating layer may be the second interlayer insulating layer **114** disposed in the display area A/A and the non-display area N/A, and the second insulating layer may be the second planarization

13

layer 116 disposed in the display area A/A and the non-display area N/A. However, the present disclosure is not limited thereto.

The spacer SP may have an undercut structure. That is, the area of the upper surface of the first spacer layer SP-1 may be smaller than the area of the lower surface of the second spacer layer SP-2. As illustrated in FIG. 4, both the first spacer layer SP-1 and the second spacer layer SP-2 have tapered shapes, and an end of the upper surface of the first spacer layer SP-1 is disposed inside an end of the lower surface of the second spacer layer SP-2, so that an undercut structure may be formed in an inward direction. However, the present disclosure is not limited thereto, and as the thickness of the first spacer layer SP-1 is smaller than that of the second spacer layer SP-2, the first spacer layer SP-1 and the second spacer layer SP-2 may have various shapes, such as an inverted tapered shape.

As the spacer SP is formed in an undercut structure between the gate driving circuit GIP and the dam DAM, the second electrode 133 disposed to overlap with the gate driving circuit GIP may be disconnected by the spacer SP. That is, the cathode electrode may be disconnected by the spacer SP. When the second electrode 133 is disconnected by the spacer SP, the cathode electrode that is not connected to the light emitting element 130 of the display area A/A no longer functions as the cathode electrode, which may be referred to as a dummy conductive layer. Accordingly, a first dummy conductive layer 133-1 is disconnected from the second electrode 133 by the spacer SP and disposed on the dam DAM. In addition, a second dummy conductive layer 133-2 is disconnected from the second electrode 133 by the spacer SP and disposed on the spacer SP. The first dummy conductive layer 133-1 and the second dummy conductive layer 133-2 may be made of the same material as the second electrode 133 of the light emitting element 130, and may be formed by the same process. In addition, since the second electrode 133 is disconnected by the spacer SP, both side surfaces of the spacer SP may be covered by the encapsulation unit 140, and the side surface of the spacer SP may be in contact with the first inorganic encapsulation layer 141 of the encapsulation unit 140.

As described above, the spacer SP may be disposed between the gate driving circuit GIP and the low potential voltage line VSS. The low potential voltage line VSS is connected to the second electrode 133 of the light emitting element 130 to supply a low potential voltage to each pixel of the display area A/A. When the second electrode 133 is disconnected by the spacer SP, the low potential voltage line VSS may not be connected to the second electrode 133 of the light emitting element 130 in a region where the spacer SP is disposed. However, since the spacer SP may be disposed only on at least one side of the left and right sides of the display area A/A like the gate driving circuit GIP, the second electrode 133 and the low potential voltage line VSS may be electrically connected to each other on at least one side of an upper side and a lower side of the display area A/A. Alternatively, when the spacer SP is disposed on only one side of the left and right sides of the display area A/A, the second electrode 133 and the low potential voltage line VSS may also be electrically connected to each other on the other side thereof.

Accordingly, the display device 100 according to the exemplary aspect of the present disclosure may improve reliability by blocking the penetration of moisture and oxygen into the display device 100 while suppressing noise caused by the inflow of the driving signal.

14

When the driving signal that may be generated in the gate driving circuit and the like is introduced into the touch routing line, the driving signal acts as noise. That is, interference occurs between the driving signal and the touch sensing signal, which may occur in the gate driving circuit, etc., to cause interference in the touch routing line, which may lead to a touch noise problem in the touch sensing unit.

The driving signal that may be generated in the gate driving circuit and the like may be partially shielded by the second electrode disposed on the upper portion of the gate driving circuit. However, the second electrode, that is, the cathode electrode may extend only to a partial region of the non-display area and may be disconnected due to a design margin. In particular, since a deposition process of the second electrode uses a process method of depositing the entire surface using an open mask without performing a photo and etching process for forming a pattern, the end of the second electrode in the non-display area has a large process tolerance and may not cover a desired location. Therefore, in the related art, when the second electrode is not deposited enough to overlap up to the touch routing line disposed at the outermost side of the plurality of touch routing lines, since the second electrode does not play a shielding role between the touch routing line and the gate driving circuit, there is a problem in that the driving signal generated in the gate driving circuit reaches the touch routing line. That is, in the related art, due to the design margin of the second electrode, there was a problem in that the driving signal is introduced into the touch routing line and causes a touch noise defect.

Accordingly, in the display device 100 according to the exemplary aspect of the present disclosure, the second electrode 133 may be designed to be extended to shield the driving signal that may be generated in the gate driving circuit GIP and the like. That is, in the deposition process of the second electrode 133, the second electrode 133 may be formed using an open mask to cover the upper portion of the dam DAM in the non-display area N/A. Accordingly, the second electrode 133 may completely cover the gate driving circuit GIP. Accordingly, in the display device 100 according to the exemplary aspect of the present disclosure, the second electrode 133 completely covers the gate driving circuit GIP, thereby minimizing the interference between the driving signal of the gate driving circuit GIP and the touch signal from occurring.

However, when the second electrode 133 extends to cover the upper portion of the dam DAM of the non-display area N/A, the second electrode 133 may be exposed to the outside of the encapsulation unit 140. As such, when the second electrode 133 is exposed to the outside of the encapsulation unit 140, there is a risk of penetration of moisture and oxygen into the display device 100.

Accordingly, in the display device 100 according to the exemplary aspect of the present disclosure, the spacer SP is disposed between the gate driving circuit GIP and the dam DAM to disconnect the second electrode 133, thereby blocking the penetration of moisture and oxygen from the outside. The spacer SP may include a first spacer layer SP-1 and a second spacer layer SP-2, and the area of the upper surface of the first spacer layer SP-1 may be smaller than that of the lower surface of the second spacer layer SP-2. Accordingly, in the process of forming the second electrode 133, the second electrode 133 may be disconnected by the spacer SP. That is, as the end of the second spacer layer SP-2 is disposed to protrude from the end of the first spacer layer SP-1, the second electrode 133 may be disposed only inside the spacer SP to cover the gate driving circuit GIP, the first

15

dummy conductive layer **133-1** may be disposed on the upper portion of the dam **DAM**, and the second dummy conductive layer **133-2** may be disposed on the upper portion of the spacer **SP**. Accordingly, in the display device **100** according to the exemplary aspect of the present disclosure, even if the moisture and oxygen are introduced along the first dummy conductive layer **133-1** or the second dummy conductive layer **133-2**, the second electrode **133** is disconnected by the spacer **SP**, thereby blocking the penetration of moisture and oxygen into the light emitting element **130** of the display area **A/A**.

FIG. **5** is a cross-sectional view of a display device according to another exemplary aspect of the present disclosure. As compared with the display device **100** of FIGS. **1** to **4**, a display device **200** of FIG. **5** is substantially the same as the display device **100** except that a spacer **SP'** and a second dummy conductive layer **S 133-3** are in plural, and thus, the duplicated description will be omitted.

Referring to FIG. **5**, a plurality of spacers **SP'** may be disposed between the gate driving circuit **GIP** and the dam **DAM** on the substrate **110**. That is, the plurality of spacers **SP'** may be disposed between the gate driving circuit **GIP** and the low potential voltage line **VSS**. Since the gate driving circuit **GIP** is disposed on at least one side of the left and right sides of the display area **A/A**, the plurality of spacers **SP'** may also be disposed on at least one side of the left and right sides of the display area **A/A**.

The plurality of spacers **SP'** includes a first spacer **SP1**, a second spacer **SP2**, and a third spacer **SP3**. The first spacer **SP1** is a spacer closest to the gate driving circuit **GIP** among the plurality of spacers **SP'**, the third spacer **SP3** is a spacer closest to the low potential voltage line **VSS** among the plurality of spacers **SP'**, and the second spacer **SP2** is a spacer disposed between the first spacer **SP1** and the third spacer **SP3**.

The plurality of spacers **SP'** may be formed of multiple layers, respectively. Specifically, the first spacer **SP1** includes a first spacer layer **SP1-1** and a second spacer layer **SP1-2**, the second spacer **SP2** includes a first spacer layer **SP2-1** and a second spacer layer **SP2-2**, and the third spacer **SP3** includes a first spacer layer **SP3-1** and a second spacer layer **SP3-2**. The first spacer layer **SP1-1** of the first spacer **SP1**, the first spacer layer **SP2-1** of the second spacer **SP2**, and the first spacer layer **SP3-1** of the third spacer **SP3** may be made of the same material as the first insulating layer disposed in the display area **A/A** and the non-display area **N/A**. The second spacer layer **SP1-2** of the first spacer **SP1**, the second spacer layer **SP2-2** of the second spacer **SP2**, and the second spacer layer **SP3-2** of the third spacer **SP3** may be made of the same material as the second insulating layer disposed in the display area **A/A** and the non-display area **N/A**. In this case, the first insulating layer disposed in the display area **A/A** and the non-display area **N/A** may be an inorganic insulating layer, and the second insulating layer may be an organic insulating layer. Specifically, the first insulating layer may be the second interlayer insulating layer **114** disposed in the display area **A/A** and the non-display area **N/A**, and the second insulating layer may be the second planarization layer **116** disposed in the display area **A/A** and the non-display area **N/A**. However, the present disclosure is not limited thereto.

The plurality of spacers **SP'** may have undercut structures. That is, the area of each of the upper surfaces of the first spacer layer **SP1-1** of the first spacer **SP1**, the first spacer layer **SP2-1** of the second spacer **SP2**, and the first spacer layer **SP3-1** of the third spacer **SP3** may be smaller than that of each of the lower surfaces of the second spacer layer

16

SP1-2 of the first spacer **SP1**, the second spacer layer **SP2-2** of the second spacer **SP2**, and the second spacer layer **SP3-2** of the third spacer **SP3**. As illustrated in FIG. **5**, the first spacer layer **SP1-1** of the first spacer **SP1**, the first spacer layer **SP2-1** of the second spacer **SP2**, and the first spacer layer **SP3-1** of the third spacer **SP3**, and the second spacer layer **SP1-2** of the first spacer **SP1**, the second spacer layer **SP2-2** of the second spacer **SP2**, and the second spacer layer **SP3-2** of the third spacer **SP3** all have tapered shapes. In addition, the ends of the upper surfaces of the first spacer layer **SP1-1** of the first spacer **SP1**, the first spacer layer **SP2-1** of the second spacer **SP2**, and the first spacer layer **SP3-1** of the third spacer **SP3** are disposed inside the ends of the lower surfaces of the second spacer layer **SP1-2** of the first spacer **SP1**, the second spacer layer **SP2-2** of the second spacer **SP2**, and the second spacer layer **SP3-2** of the third spacer **SP3**. Accordingly, the undercut structure may be formed in an inward direction, but the present disclosure is not limited thereto. The thicknesses of the first spacer layer **SP1-1** of the first spacer **SP1**, the first spacer layer **SP2-1** of the second spacer **SP2**, and the first spacer layer **SP3-1** of the third spacer **SP3** are smaller than those of the second spacer layer **SP1-2** of the first spacer **SP1**, the second spacer layer **SP2-2** of the second spacer **SP2**, and the second spacer layer **SP3-2** of the third spacer **SP3**. In addition, the first spacer layer **SP1-1** of the first spacer **SP1**, the first spacer layer **SP2-1** of the second spacer **SP2**, and the first spacer layer **SP3-1** of the third spacer **SP3**, and the second spacer layer **SP1-2** of the first spacer **SP1**, the second spacer layer **SP2-2** of the second spacer **SP2**, and the second spacer layer **SP3-2** of the third spacer **SP3** may have various shapes such as an inversed tapered shape.

As the plurality of spacers **SP'** is formed in undercut structures between the gate driving circuit **GIP** and the dam **DAM**, the second electrode **133** disposed to overlap with the gate driving circuit **GIP** may be easily disconnected by the plurality of spacers **SP'**. Accordingly, the first dummy conductive layer **133-1** is disconnected from the second electrode **133** by the spacer **SP'** and disposed on the dam **DAM**. In addition, a plurality of second dummy conductive layers **133-3** is disconnected from the second electrode **133** by the spacer **SP'** and disposed on the plurality of spacers **SP'**. The first dummy conductive layer **133-1** and the plurality of second dummy conductive layers **133-3** may be made of the same material as the second electrode **133** of the light emitting element **130**, and may be formed by the same process.

In the display device **200** according to another exemplary aspect of the present disclosure, the second electrode **133** may be designed to be extended to shield the driving signal that may be generated in the gate driving circuit **GIP**. That is, in the deposition process of the second electrode **133**, the second electrode **133** may be formed using an open mask so that the second electrode **133** may also cover the upper portion of the dam **DAM** in the non-display area **N/A**. Accordingly, the second electrode **133** may completely cover the gate driving circuit **GIP**. Accordingly, in the display device **200** according to another exemplary aspect of the present disclosure, the second electrode **133** completely covers the gate driving circuit **GIP**, thereby minimizing interference between the driving signal of the gate driving circuit **GIP** and the touch signal from occurring.

In addition, in the display device **200** according to another exemplary aspect of the present disclosure, the plurality of spacers **SP'** is disposed between the gate driving circuit **GIP** and the dam **DAM** to disconnect the second electrode **133**, thereby blocking the penetration of moisture and oxygen

17

from the outside. Since each of the plurality of spacers SP' has an inverted tapered shape, the second electrode 133 may be more reliably disconnected. Accordingly, in the display device 200 according to another exemplary aspect of the present disclosure, even if the moisture and the oxygen are introduced along the first dummy conductive layer 133-1 or the plurality of second dummy conductive layers 133-3, the second electrode 133 is disconnected by the plurality of spacers SP', thereby blocking the penetration of moisture and oxygen into the light emitting element 130 of the display area A/A.

The exemplary aspects of the present disclosure may also be described as follows:

According to an aspect of the present disclosure, there is provided a display device. The display device includes a substrate including a display area and a non-display area surrounding the display area. The display device further includes a gate driving circuit disposed on the substrate and in the non-display area. The display device further includes a light emitting element disposed in the display area and on the substrate and including a cathode electrode. The display device further includes a dam disposed on the substrate and in the non-display area, the dam further outside the substrate than the gate driving circuit on the substrate. The display device further includes a first dummy conductive layer disposed to cover the dam and made of the same material as the cathode electrode. The display device further includes a spacer disposed between the gate driving circuit and the dam on the substrate. The cathode electrode extends from the display area to the non-display area to overlap with the gate driving circuit.

The display device may further include an encapsulation unit covering the gate driving circuit, the dam and the light emitting element. The display device may further include a touch sensing unit disposed on the encapsulation unit. The touch sensing unit may include a touch electrode disposed in the display area and a touch routing line connected to the touch electrode and disposed in the non-display area. The touch routing line may overlap with the cathode electrode.

The cathode electrode and the first dummy conductive layer may be disconnected by the spacer.

The display device may further include a first insulating layer disposed in the display area and the non-display area on the substrate. The display device may further include a second insulating layer disposed on the first insulating layer. The spacer may include a first spacer layer made of the same material as the first insulating layer and a second spacer layer made of the same material as the second insulating layer. The spacer may have an undercut structure.

An area of an upper surface of the first spacer layer may be smaller than that of a lower surface of the second spacer layer.

The first insulating layer may be an inorganic insulating layer. The second insulating layer may be an organic insulating layer.

A side surface of the spacer may be in contact with an encapsulation unit.

The display device may further include a second dummy conductive layer disposed on the spacer and made of the same material as the cathode electrode.

The gate driving circuit may be disposed on at least one side of a left side and a right side of the display area.

The display device may further include a low potential voltage line electrically connected to the cathode electrode. The spacer may be disposed between the gate driving circuit and the low potential voltage line.

18

The cathode electrode and the low potential voltage line may be electrically connected to each other in at least one side of an upper side and a lower side of the display area.

The low potential voltage line may be formed on the same layer as the gate driving circuit.

The first spacer layer and the second spacer layer may have inverted tapered shapes.

The spacer may include one or more spacer.

Although the exemplary aspects of the present disclosure have been described in detail with reference to the accompanying drawings, the present disclosure is not limited thereto and may be embodied in many different forms without departing from the technical concept of the present disclosure. Therefore, the exemplary aspects of the present disclosure are provided for illustrative purposes only but not intended to limit the technical concept of the present disclosure. The scope of the technical concept of the present disclosure is not limited thereto. Therefore, it should be understood that the above-described exemplary aspects are illustrative in all aspects and do not limit the present disclosure. The protective scope of the present disclosure should be construed based on the following claims, and all the technical concepts in the equivalent scope thereof should be construed as falling within the scope of the present disclosure.

What is claimed is:

1. A touch display device comprising:

a flexible substrate including a display area and a non-display area adjacent to one or more side surfaces of the display area;

a light emitting element disposed on the flexible substrate in the display area and including a first electrode and a second electrode;

a first inorganic encapsulation layer on the light emitting element;

an organic encapsulation layer on the first inorganic encapsulation layer;

a second inorganic encapsulation layer on the organic encapsulation layer;

a touch insulating layer on the second inorganic encapsulation layer;

a gate driving circuit disposed on the flexible substrate in the non-display area; and

a plurality of dams disposed on the flexible substrate in the non-display area,

wherein the second electrode extends from the display area to the non-display area and is disconnected between the display area and the dams,

wherein a side surface of the second electrode is in contact with the first inorganic encapsulation layer,

wherein the organic encapsulation layer is disposed between at least two dams among the plurality of dams,

wherein the second inorganic encapsulation layer and the touch insulating layer extend in a direction opposite to the display area and cover at least one of the plurality of dams, and

wherein the plurality of dams disposed further outside the flexible substrate than the gate driving circuit.

2. The touch display device of claim 1, further comprising a touch sensing unit disposed on the touch insulating layer.

3. The touch display device of claim 1, further comprising:

a thin film transistor disposed on the flexible substrate in the display area and including an active layer, a gate electrode, a source electrode, and a drain electrode; and an interlayer insulating layer disposed on the gate electrode,

19

wherein an end of the second electrode is in contact with the interlayer insulating layer in the non-display area.

4. The touch display device of claim 1, further comprising:

a first dummy conductive layer covering the plurality of dams in the non-display area and made of a same material as the second electrode; and

at least one spacer disposed on the flexible substrate between the gate driving circuit and the dams in the non-display area.

5. The touch display device of claim 4, wherein the second electrode and the first dummy conductive layer are disconnected by the spacer.

6. The touch display device of claim 2, wherein the touch sensing unit includes:

a touch electrode disposed in the display area, and

a touch routing line disposed in the non-display area and connected to the touch electrode,

wherein the touch routing line overlaps with the second electrode.

7. The touch display device of claim 4, further comprising:

a first insulating layer disposed in the display area and the non-display area on the flexible substrate; and

a second insulating layer disposed on the first insulating layer,

wherein the spacer includes a first spacer layer made of a same material as the first insulating layer and a second spacer layer made of a same material as the second insulating layer, and the spacer has an undercut structure,

wherein the light emitting element disposed on the second insulating in the display area.

20

8. The touch display device of claim 7, wherein an area of an upper surface of the first spacer layer is smaller than that of a lower surface of the second spacer layer.

9. The touch display device of claim 7, wherein the first insulating layer includes an inorganic insulating layer, and the second insulating layer includes an organic insulating layer.

10. The touch display device of claim 5, wherein a side surface of the spacer is in contact with the first inorganic encapsulation layer.

11. The touch display device of claim 5, further comprising a second dummy conductive layer disposed on the spacer and made of a same material as the second electrode.

12. The touch display device of claim 5, wherein the gate driving circuit is disposed on at least one side of a left side and a right side of the display area.

13. The touch display device of claim 5, further comprising a low potential voltage line electrically connected to the second electrode,

wherein the spacer is disposed between the gate driving circuit and the low potential voltage line.

14. The touch display device of claim 13, wherein the second electrode and the low potential voltage line are electrically connected to each other in at least one side of an upper side and a lower side of the display area.

15. The touch display device of claim 13, wherein the low potential voltage line is formed on a same layer as the gate driving circuit.

16. The touch display device of claim 7, wherein at least one of the first spacer layer and the second spacer layer have an inversed tapered shape.

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